Fundamentals 30

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Problem 30.1

Part (1)

Define

$$f_n(x) = \max(0, 1 - |x - n|).$$

Then this satisfies the constraints.

Part (2)

Let $x \in \mathbb{R}$ and $N > x + 1$. If $n \geq N$, then $f_n(x) = 0$ since $|x - n| = n - x \geq N - x > 1$. So $f_n(x) \to 0$ for all x.

Part (3)

Let $\varepsilon < 1$. Then, since $\sup\{d(f(x), g(x)) : x \in X\} \geq 1$, $\rho(f, g)$ will never be $< \varepsilon$. So f_n doesn't converge in the uniform topology.

Part (4)

Since (f_n) converges pointwise but not uniformly, the uniform and pointwise topologies aren't equivalent.

Part (5)

Yes, f_n converges in the CC topology. To show this, let $C \subseteq \mathbb{R}$ and $N > \sup(C) + 1$. Then by similar logic as before, if $x \in C$ then $|x - n| > 1$, so $f_n(x) = 0$, and therefore $f_n \to 0$ in the CC topology.

Problem 30.2

We prove the axioms of a topological basis:

1. Let $f \in Y^X$. Choose any $C \subseteq X$ and $\varepsilon > 0$. Then $\rho(f, f) = 0 < \varepsilon$, so $f \in B_C(f, \varepsilon)$.

2. Let $B_1 = B_{C_1}(f_1, \varepsilon_1)$ and $B_2 = B_{C_2}(f_2, \varepsilon_2)$, and $g \in B_1 \cap B_2$. Let $\delta_1 = \rho(f_1, g) < \varepsilon_1$ and $\delta_2 = \rho(f_2, g) < \varepsilon_2$. Then let $C_3 = C_1 \cup C_2$ (which is compact), $f_3 = g$, and $\varepsilon_3 = \min(\varepsilon_1 - \delta_1, \varepsilon_2 - \delta_2) > 0$. If $h \in B_{C_3}(g, \varepsilon_3)$ then, for each $i = 1, 2$ and $x \in C_i$,

$$d(f_i(x), h(x)) \leq d(f_i(x) + g(x)) + d(g(x), h(x)) < \delta_i + \varepsilon_3 \leq \varepsilon_i.$$

So, for each $i = 1, 2$, $h \in B_{C_i}(f_i, \varepsilon_i)$.

Problem 30.3

First, assume $f_n \to f$ in the CC topology. For every $B_C(f, \varepsilon)$, there is N such that for all $n \geq N$, if $f_n \in B_C(f, \varepsilon)$ then $\rho(f_n, f) < \varepsilon$. This is exactly the same statement as (f_n) converges uniformly to f on C.

Now, assume that for every $C \subseteq X$, $\rho(f_n, f) \to 0$. Then, given any $B_C(f, \varepsilon)$, by uniform convergence, there is N such that for all $n \geq N$, $\rho(f_n, f) < \varepsilon$, so $f_n \in B_C(f, \varepsilon)$. So $f_n \to f$ in the CC topology.

Problem 30.4

We have that

$$(f|_C)^{-1}(V) = \{x \in C : f|_C(x) \in V\} = \{x \in C : f(x) \in V\},$$

and

$$f^{-1}(V) \cap C = \{x \in X : f(x) \in V \text{ and } x \in C\} = \{x \in C : f(x) \in V\}.$$

Therefore, the two sets are equal.

Problem 30.7

Let $x_n \to x$ in X where (x_n) is a sequence in C. Since C is closed, $U = X \backslash C$ is open. For contradiction, assume $x \notin C$, so that $x \in U$. Since U is open, U is a neighborhood of x. But since $x_n \to x$, there exists N such that for all $n \geq N$, $x_n \in U$. This contradicts the fact that all x_n are in C, so $x \in C$.

Problem 30.8

Let $\varepsilon > 0$. Then for any n, if $x \geq n\varepsilon$, $f_n(x) \geq \varepsilon$, so it doesn't converge uniformly. Now, let $C \subseteq \mathbb{R}$ be closed and bounded and $\varepsilon > 0$. If we let $N > \frac{\sup(C)}{\varepsilon}$, then if $n \geq N$, then for all $x \in C$,

$$f_n(x) = \frac{x}{n} \leq \frac{x}{N} < \frac{x\varepsilon}{\sup(C)} < \varepsilon$$

since $x < \sup(C)$. So (f_n) converges in the CC topology but not uniformly.

Problem 30.9

The sequence (f_n) converges to $\frac{x}{(1-x)^2}$ on $(-1,1)$ in the CC topology but not the uniform topology.

Let $C \subseteq (-1,1)$ be compact. Then there exists $r < 1$ such that $C \subseteq [-r,r]$. For any $x \in C$,

$$|f(x) - f_n(x)| = \left| \sum_{k=n+1}^{\infty} kx^k \right| \leq \sum_{k=n+1}^{\infty} k|x|^k \leq \sum_{k=n+1}^{\infty} kr^k.$$

Since $|r| < 1$, this sum converges to 0 as $n \to \infty$. So the sum converges for all $x \in C$.

We have that

$$\lim_{n\to 1} f(x) - f_n(x) = \lim_{n\to\infty} \sum_{k=n+1}^{\infty} kx^k = \infty,$$

and

$$\sup\{f(x) - f_n(x)\} \geq \lim_{n\to 1} f(x) - f_n(x),$$

so (f_n) is not uniformly convergent.